UNIT 2: SHAPES AND REGIONS

SESSION 8: BOUNDARY DESCRIPTORS

SLO 2: APPLY BOUNDARY DESCRIPTORS TO EXTRACT MEANINGFUL FEATURES FOR SHAPE ANALYSIS AND RECOGNITION.

Descriptor Decision Lab □

A vision system extracted the following boundary information from objects.

Case	Boundary Observation	What Feature Can Be Extracted?
1	Boundary has 4 sharp corners	Corner count / vertex detection identifies a polygon (e.g., 4 corners = quadrilateral/square).
2	Boundary is smooth with no corners	Zero curvature / no corner points indicates a circular or elliptical (rounded) shape.
3	Boundary length is very large	High perimeter relative to enclosed area indicates an irregular or complex (highly convoluted) boundary shape.

Question:

Which descriptor property helps identify shape complexity?

Answer: Compactness (perimeter² / area) – a longer, more convoluted boundary relative to the enclosed area gives a higher compactness value, indicating greater shape complexity.

Activity 2 – Error Detective 🔎

A computer vision system produced the following shape recognition results.

Detected Feature	System Prediction
Smooth continuous boundary	Square
Four equal edges	Circle
Three corners	Triangle

Tasks

1. **Identify the incorrect classifications.**

   Row 1 (Smooth continuous boundary → predicted Square) is incorrect, and Row 2 (Four equal edges → predicted Circle) is incorrect. Row 3 (Three corners → predicted Triangle) is correct.

2. **Correct the actual shapes based on boundary descriptors.**

   Smooth continuous boundary → actually a Circle. Four equal edges → actually a Square. Three corners → correctly a Triangle.

3. **Explain which boundary feature was misinterpreted.**

   The system confused the corner-count descriptor with the smoothness descriptor: it read a smooth, corner-free boundary as belonging to a shape with corners (Square) and a boundary with sharp, equal edges as having no corners (Circle) – i.e., it swapped which feature (curvature/corner presence vs. edge count) signals which shape.